Airbnb Data Analysis

Predicting nightly price in New York City using public Airbnb listings data

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IS 4487 | Group 5

Business Analytics Case Study

NYC short-term rental pricing

Group 5

Airbnb price prediction

Public data
Model comparison
Business recommendations

Final deliverable

PowerPoint + notebook

based on your project work

Executive Summary

What the project is solving and what the models found

Business problem

Airbnb hosts in New York City need a practical way to estimate nightly prices from public listing characteristics. Pricing too high can reduce bookings; pricing too low leaves revenue on the table. The project asks which listing features matter most and how well a model can predict price.

Best model

Random Forest Regressor was the strongest full-set model.

Typical test-set error

About $79 MAE and $282 RMSE on the full test set.

Main caution

Extreme luxury listings drove much of the error, so the model works best for standard listings rather than very high-priced outliers.

Full-test-set Random Forest performance: RMSE $282.32 | MAE $78.76 | R^2 0.54

Data Understanding

Inside Airbnb NYC listings data and the scope of the analysis

Dataset source

Inside Airbnb – New York City listings.csv and the accompanying data dictionary. The raw file contained 30,259 listings and about 90 variables describing listing characteristics, host information, reviews, pricing, availability, and location.

Key fields used

Location, room type, host status, reviews, amenities, availability, and engineered geographic and review-history features. The target variable is nightly price, modeled after log transformation.

After preparation

The cleaned modeling set contained 21,512 rows and 96 columns. Missing prices were removed, implausible low prices were filtered out, and the remaining features were transformed for model use.

Raw data -> cleaning -> engineered features -> model-ready dataset

Data Preparation

Cleaning and feature engineering choices that shaped the models

1. Clean the target

Removed rows with missing price values and filtered implausible prices below $10. Empty columns were dropped.

2. Engineer features

Created log_price, availability rates, near-term availability gap, distance to Midtown, grouped property types, review-history variables, and amenity flags/counts.

3. Model-ready data

Used a train/test split, median imputation for numeric fields, "Unknown" for categorical gaps, one-hot encoding, and scaling for the linear model only.

Important choice: all preprocessing was fit on training data only to prevent leakage into the test set.

Modeling Approach

A baseline plus two predictive models, with a tuned Random Forest

Baseline

Predicts the training-set mean price for every listing. Sets the minimum benchmark.

Linear Regression

Interpretable benchmark that predicts log(price) and shows direction and size of feature effects.

Random Forest

Captures non-linear patterns and interactions in listing characteristics.

Tuned Random Forest

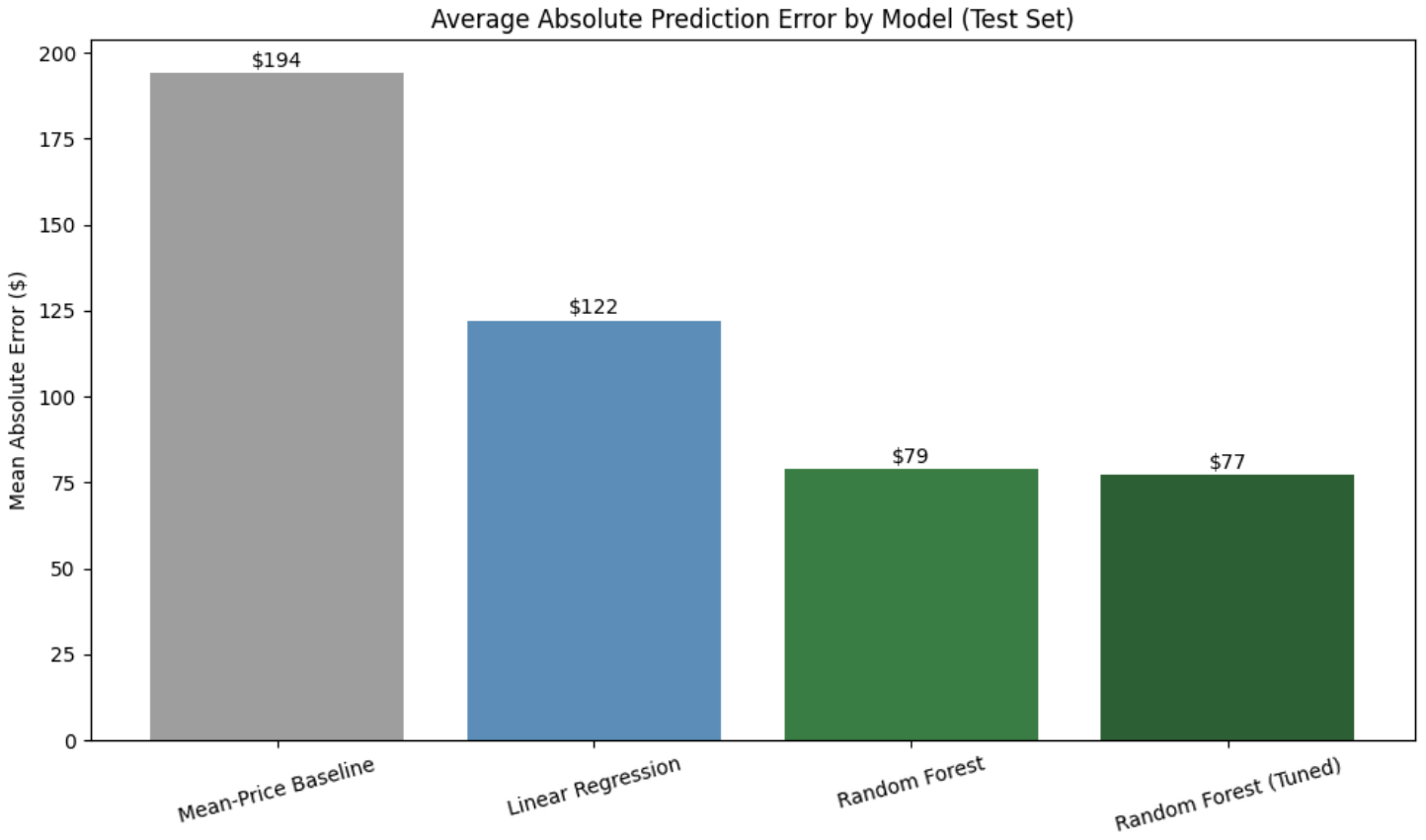
Randomized search on a 5,000-row sample; retrained with the best parameters.

Validation: train/test split + 2-fold randomized search for the tuned Random Forest

All models were trained on log(price) and converted back to dollars for business interpretation.

Model Performance Comparison

Random Forest outperformed the baseline and the linear model on the full test set



Model	RMSE	MAE	R^2
Mean baseline	$415.73	$194.03	-0.00
Linear Regression	$1,376.34	$122.18	-9.97
Random Forest	$282.32	$78.76	0.54
Random Forest (Tuned)	$288.50	$77.05	0.52

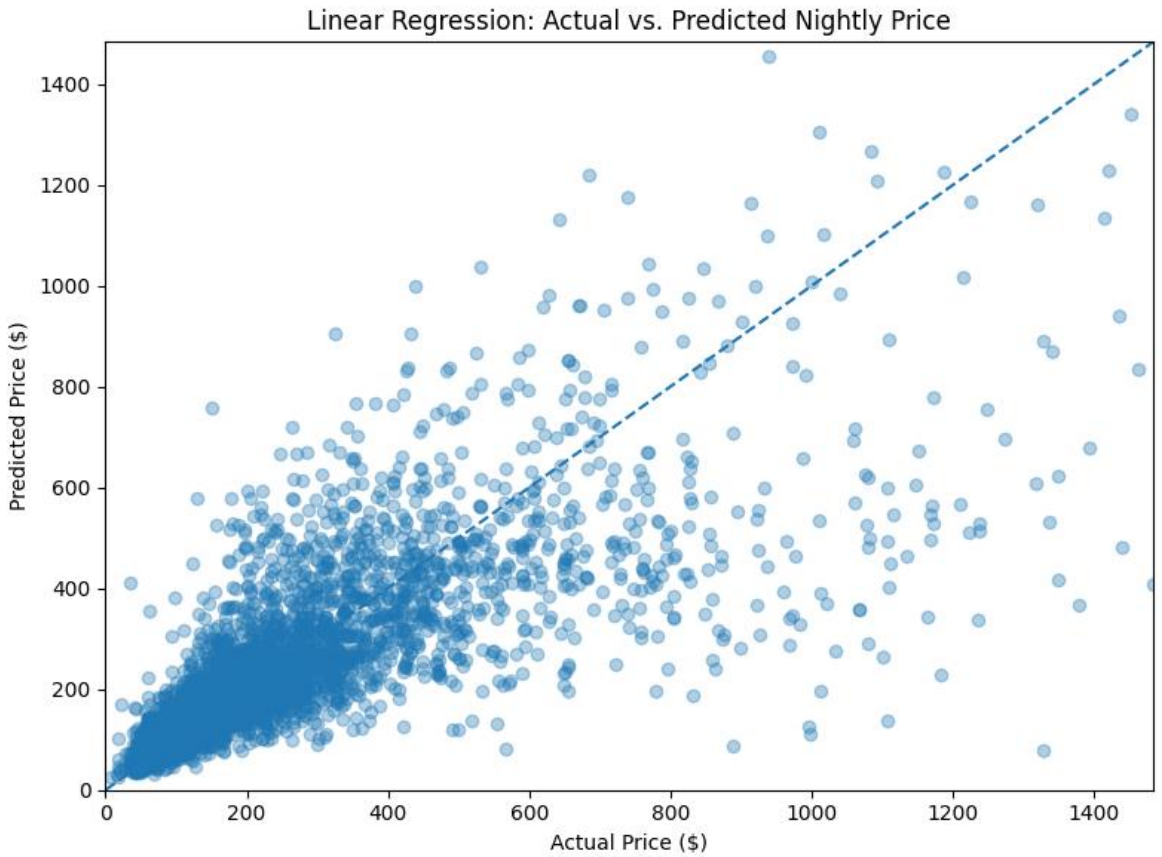
Key takeaway

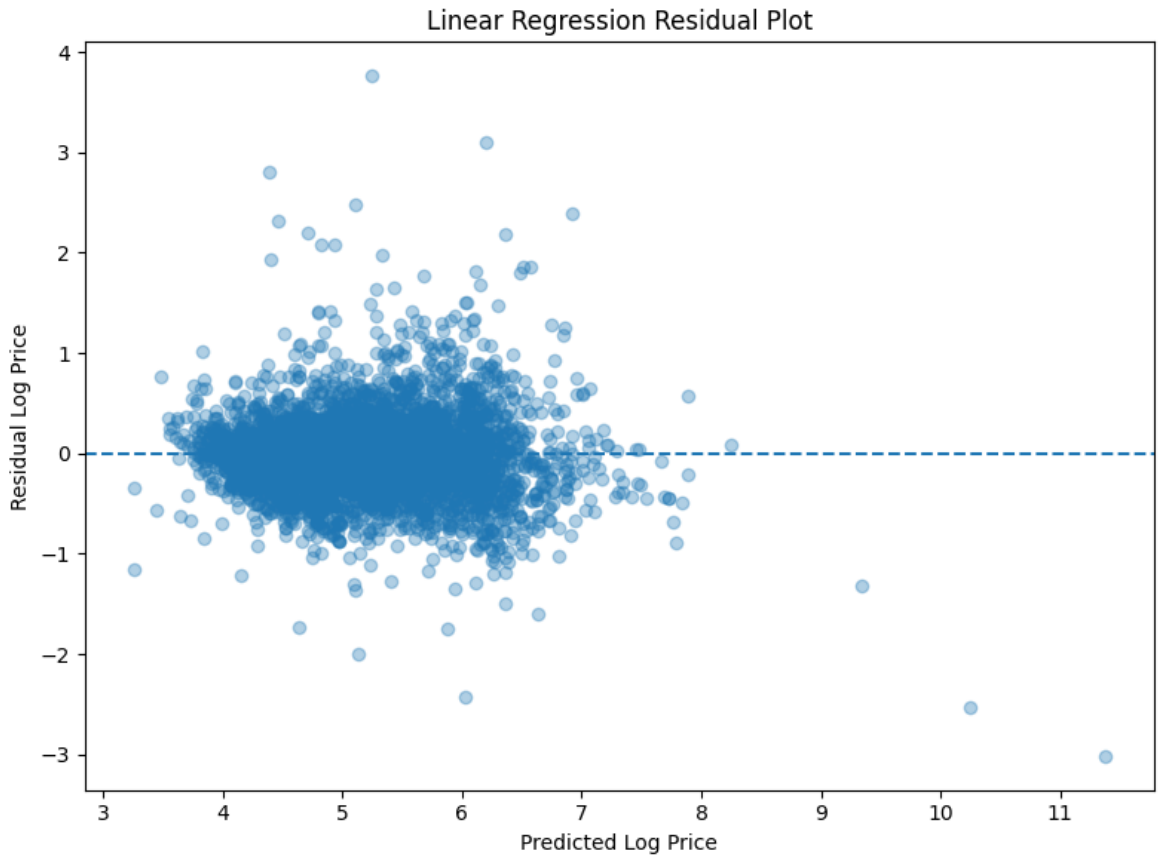
Random Forest cut the average absolute error from about $194 to about $79 on the full test set.

That makes it the best choice for rough pricing guidance.

Model Diagnostics

The linear model follows the broad pattern but struggles with extreme prices

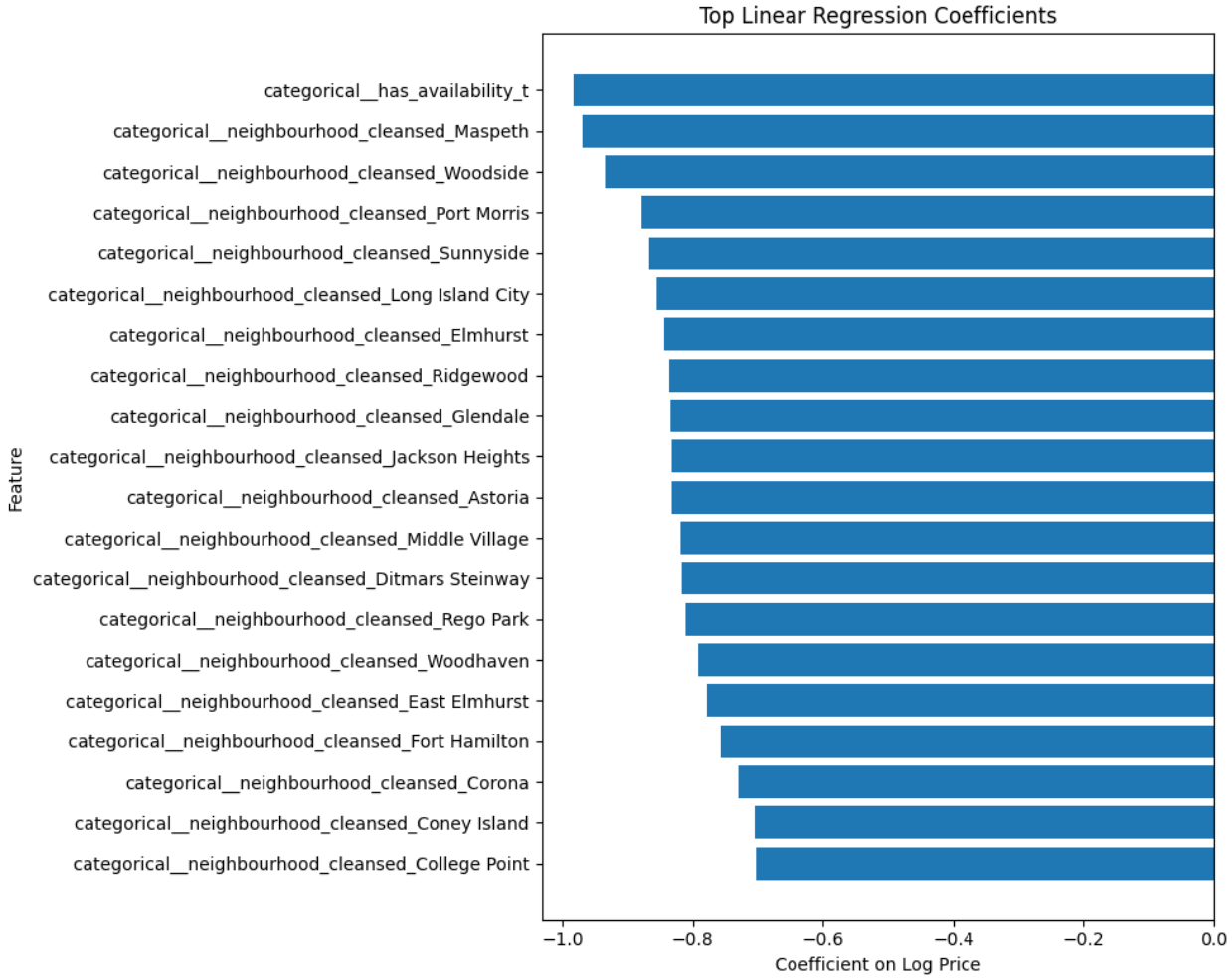




Interpretation: the model works best in the middle of the price range, but high-end outliers widen the spread and weaken fit on the full scale.

What the Linear Model Suggests

Coefficients help explain price direction even when prediction accuracy is weaker



Location signal

Several Queens neighborhood indicators carried large negative coefficients relative to the reference category, showing that location was a major price driver.

Availability signal

Availability also appeared in the coefficient ranking, suggesting a relationship between how open a listing is and its predicted price.

How to use it

Use Linear Regression to explain which features move the price up or down, but use Random Forest for the final pricing estimate.

Recommendations, Ethics, and Limitations

How to use the model responsibly

Recommendations

- Use the Random Forest model as a rough pricing tool for standard listings.
- Treat predictions above about $1,400 per night with caution because outliers were less stable.
- Combine model output with human judgment and competitive market checks.

Ethics and limitations

- Historical data can carry bias and underrepresent certain groups.
- Removing 8,744 missing-price rows may create selection bias.
- Listing and host data should be handled carefully for privacy.
- Predictions should support decisions, not replace them.

Best use case: decision support for hosts and analysts, not fully automated pricing.

References

Sources included in the project notebook

Inside Airbnb

https://insideairbnb.com/get-the-data/ | listings.csv and the NYC data dictionary

Industry context

Hotel Management Network: Airbnb vs. hotels article used for hospitality comparison context

Additional reference from notebook

Times of India article on property-owner notifications and potential tax pressure

Thank you

Questions?